Computer Networks Lab

Cisco Packet Tracer — Assignment 6

All practicals must be completed inside **Cisco Packet Tracer**.
This assignment requires **two screenshots** as submission.

Topic: Extended Access Control Lists (ACL)

How to Submit

- Complete the practical entirely inside **Cisco Packet Tracer**. No other tool is accepted.
- Save your file as `RollNo_Set6.pkt` (e.g. `20221234_Set6.pkt`) and submit it.
- Along with the file, submit exactly **two screenshots** as described at the end of the set.
- Screenshots must be clear and show all device labels and relevant details.

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Scenario

Fortuna Bank has three departments connected through a central router: **HR**, **Finance**, and **IT**. The IT department hosts a **Payroll Server** that provides the internal payroll web portal where salary records are viewed, and it also handles the salary-slip email that is sent to employees.

Recently, the head officer, Mr. Niraj Mehta, discovered that HR staff have been opening the payroll portal and browsing salary records that are not theirs, and have also been reading salary-slip emails addressed to other employees. He has decided that **HR must be cut off from every service on the Payroll Server that handles payroll data**, not just the portal. However, HR still needs to **ping the Payroll Server** to confirm it is online before each payroll run, and all other traffic must continue to work normally. The **Finance** department keeps full access to all of the server's services.

Device	IP Address	Default Gateway
Router Gi0/0 (HR)	192.168.10.1	—
HR-PC1	192.168.10.10	192.168.10.1
HR-PC2	192.168.10.20	192.168.10.1
Router Gi0/1 (Finance)	192.168.20.1	—
Fin-PC1	192.168.20.10	192.168.20.1
Fin-PC2	192.168.20.20	192.168.20.1
Router Gi0/2 (IT)	192.168.30.1	—
Payroll-Server	192.168.30.10	192.168.30.1
Subnet Mask: `255.255.255.0` on all devices		

Your Task

- Build the topology: one router with three interfaces, one switch per department (HR, Finance, IT), two PCs each in HR and Finance, and the Payroll Server on the IT side.
- Assign all IP addresses and default gateways from the table, and make sure every router interface is active.
- Enable the services the Payroll Server must provide to the bank (the payroll web portal and the salary-slip email service), and set up an email account on the HR and Finance PCs so the services can be tested.
- Before configuring any ACL, verify that an HR PC can:
  (a) open the payroll portal (`http://192.168.30.10`),
  (b) send and receive salary-slip email through the server, and
  (c) ping the Payroll Server.
- Configure an **Extended ACL** (number **100–199**) on the router that blocks HR from every service on the Payroll Server that handles payroll data, while permitting everything else, including ping from HR.
- Apply the ACL on the correct interface and in the correct direction so that the rules are

checked as HR traffic enters the router, without affecting Finance's access in any way.

- Re-test connectivity:
  (a) the HR PC must **fail** to load the payroll portal,
  (b) the HR PC must **fail** to send and receive salary-slip email,
  (c) the HR PC's ping to the Payroll Server must still **succeed**, and
  (d) the Finance PC must still **load** the portal and **send and receive** email.

Submission

Submit the following **two screenshots**:

1. The completed **topology** in Packet Tracer Logical View with all devices connected and IP labels visible, along with the router's configuration showing the Extended ACL rules and the interface they are applied to.
2. A single screenshot showing the HR PC's **Web Browser** failing to load the payroll portal, its **Email** client failing to send or receive, and its **ping** to the Payroll Server succeeding, together with a Finance PC's **Web Browser** loading the portal successfully.

Note: A **Standard ACL** can only check the source address, so it would also block HR's ping and every other service. An **Extended ACL** can inspect the protocol, the destination address, and the destination port number, which allows much more precise control. Because it filters this precisely, Cisco's best practice is to place an extended ACL **as close to the source** of the traffic as possible. Remember that ACL rules are checked from top to bottom, the first match wins, and every ACL ends with an implicit deny.